Roll No. 24L-0781 Section 4E

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Database Systems	Course Code:	CS 2001
Program:	BS(CS)	Semester:	Spring 2026
Duration:	10 Minutes	Total Marks:	10
Paper Date:	11 Feb 2026	Exam	Quiz 1
Section:	E		

15/15

Instruction/Notes: Solve the exam on this question paper.

Question: Consider the following banking

Accounts

ID	Holder_ID	Account_Type	Balance
10	1	Current	1000
20	2	Saving	2000
30	1	Saving	3000

AccountHolders

HID	Name
1	John
2	Jane
3	Alex

Transactions

Trans_ID	AccountID	Type	Amount	Date
100	10	Deposit	3000	9/2/2026
101	20	Withdraw	100	3/2/2026
102	40	Deposit	500	8/10/2025

Write SQL and RA queries for the following tasks:

a) Give the AccountID and the Account holder name for accounts with a balance less than 2000
b) Find the ID of account holders who hold both the Current account and the Savings Account
c) Find the Account for which no "Deposit" transaction is performed on 10/2/2026

a) Select Account ID, ~~Account Holder-ID~~ Name
FROM (Accounts Join Account Holders ON HolderID = HID)
Join Transactions ON ID = AccountID
where Balance ~~>~~ < 2000

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RA → $\pi_{\text{Account ID, Name}}(\sigma_{\text{Bal} < 2000}((\text{Account} \bowtie_{\text{Holder ID} = \text{HID}} \text{Account Holders}) \bowtie_{\text{ID} = \text{Account ID}} \text{Transactions}))$

B) Select ~~ID~~ Holder-ID
FROM Accounts
where Account Type = "Current"
INTERSECTION
Select ~~ID~~ Holder-ID
FROM Accounts
where AccountTYPE = "Saving"

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C) Select Account ID
FROM Transactions
~~where Type~~
Except
Select AccountID
FROM Transactions
where Type = "Deposit"
AND Date like
"10/02/2026"

5

C)

RA $\pi_{\text{Account ID}}$ (Transactions $\overset{\text{Except}}{-}$ Transactions $\sigma_{\text{Type = Deposit AND Date = "10/02/2026"}}$)

B) RA -> $\pi_{\text{Holder-ID}}$ ($\sigma_{\text{Accounts.TYPE = "Current"}}$ $\cap$ $\sigma_{\text{Accounts.TYPE = "saving"}}$)